(13) $\int \frac{e^x(x^2\tan^{-1}x+\tan^{-1}x+1)}{x^2+1}\,dx =$

(1) $e^x\tan^{-1}(x+1)+c$ (2) $\tan^{-1}(e^x)+c$ (3) $e^x\frac{(\tan^{-1}x)^2}{2}+c$ (4) $e^x\tan^{-1}x+c$

(14) $\int \frac{x^2+\cos^2 x}{x^2+1}\operatorname{cosec}^2 x\,dx =$

(1) $\cot x+\sin^{-1}x+c$ (2) $-\cot x+\tan^{-1}x+c$

(3) $-\tan x+\cot^{-1}x+c$ (4) $-\cot x-\tan^{-1}x+c$

(15) $\int x^2\cos x\,dx =$

(1) $x^2\sin x+2x\cos x-2\sin x+c$ (2) $x^2\sin x-2x\cos x-2\sin x+c$

(3) $-x^2\sin x+2x\cos x+2\sin x+c$ (4) $-x^2\sin x-2x\cos x+2\sin x+c$

(16) $\int \sqrt{\frac{1-x}{1+x}}dx =$

(1) $\sqrt{1-x^2}+\sin^{-1}x+c$ (2) $\sin^{-1}x-\sqrt{1-x^2}+c$

(3) $\log|x+\sqrt{1-x^2}|-\sqrt{1-x^2}+c$ (4) $\sqrt{1-x^2}+\log|x+\sqrt{1-x^2}|+c$

(17) $\int \frac{dx}{e^x-1}=$

(1) $\log|e^x|-\log|e^x-1|+c$ (2) $\log|e^x|+\log|e^x-1|+c$

(3) $\log|e^x-1|-\log|e^x|+c$ (4) $\log|e^x+1|-\log|e^x|+c$

(18) $\int e^{-4x}\cos x\,dx=$

(1) $\frac{e^{-4x}}{17}[4\cos x-\sin x]+c$ (2) $\frac{e^{-4x}}{17}[-4\cos x+\sin x]+c$

(3) $\frac{e^{-4x}}{17}[4\cos x+\sin x]+c$ (4) $\frac{e^{-4x}}{17}[-4\cos x-\sin x]+c$

(19) $\int \frac{\sec^2 x}{\tan^2 x-1}dx=$

(1) $2\log\left|\frac{1-\tan x}{1+\tan x}\right|+c$ (2) $\log\left|\frac{1+\tan x}{1-\tan x}\right|+c$

(3) $\frac{1}{2}\log\left|\frac{\tan x+1}{\tan x-1}\right|+c$ (4) $\frac{1}{2}\log\left|\frac{\tan x-1}{\tan x+1}\right|+c$

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